

Outdoor WiFi + Camera Installation

Sand Volleyball Courts - Design & Placement Report

Manor Lanes / WNY Social Sports - 150 Grand Island Blvd

Rev. 5 - UDB-Switch architecture; courts + patio cameras; measured distances + full specs

Site	3 outdoor sand volleyball courts + 2 patios (P1, P2)
Platform	Ubiquiti UniFi (Network + Protect)
Installer	Brian Russo (owner-installer)
Controller / NVR	CloudKey+ (SSD) + Network Video Recorder Pro + 16 TB HDD (existing)
Key constraints	No trenching; no new cable across sand/drainage; cold-weather (Buffalo NY)
Backhaul method	UniFi UDB-Switch: WiFi-7 wireless-uplink switches at each pole (no outdoor cable)
Access points	3x U6 Mesh Pro - relocate 2 off building wall + add 1 new
Net new hardware	approx. \$1,500 - \$2,000 (most network gear repositioned, not repurchased)

Design summary

Each court pole carries a UniFi UDB-Switch - a WiFi-7 switch with a built-in wireless uplink. It backhauls to the UniFi WiFi over the air (no cable across the sand or drainage) AND supplies local PoE to that pole's cameras and access point. A U6 Mesh Pro re-mounted on the building (off the concrete wall) is the wired uplink anchor; the center-pole UDB-Switch links to it, and the Court 3 UDB-Switch links (relay) to the center-pole AP. The three U6 Mesh Pro APs are repositioned from the failed concrete-wall mounting onto poles, where coverage radiates over the sand. Court cameras: 3 fixed G6 Bullets (one per court). Patio cameras (wired to the rack, not via the backhaul): 1 G6 PTZ (pole-mounted - covers the 2nd patio + zoom/auto-track) + 2 G6 Bullets on the main-patio back wall. The AI PTZ and a G6 180 remain on the parking lot.

1. Background & Existing Conditions

Problem identified

- Two U6 Mesh Pro APs were wall-mounted on the concrete back of the building.
- Heat-map survey showed strong signal (-30 to -50 dBm) only at the wall; the entire court surface measured -70 to -80 dBm (marginal-to-unusable), failing under league crowds.
- Root cause: omnidirectional APs flat on concrete radiate left/right along the wall and reflect/absorb into the concrete - signal does not project across the open sand.

Fix

- Reposition the U6 Mesh Pro APs off the wall onto field poles so coverage radiates over the sand.
- Use UDB-Switch wireless-uplink switches for backhaul - no cable across the sand/drainage; each pole gets local PoE for its cameras + AP.

Existing UniFi equipment to be REUSED (no purchase)

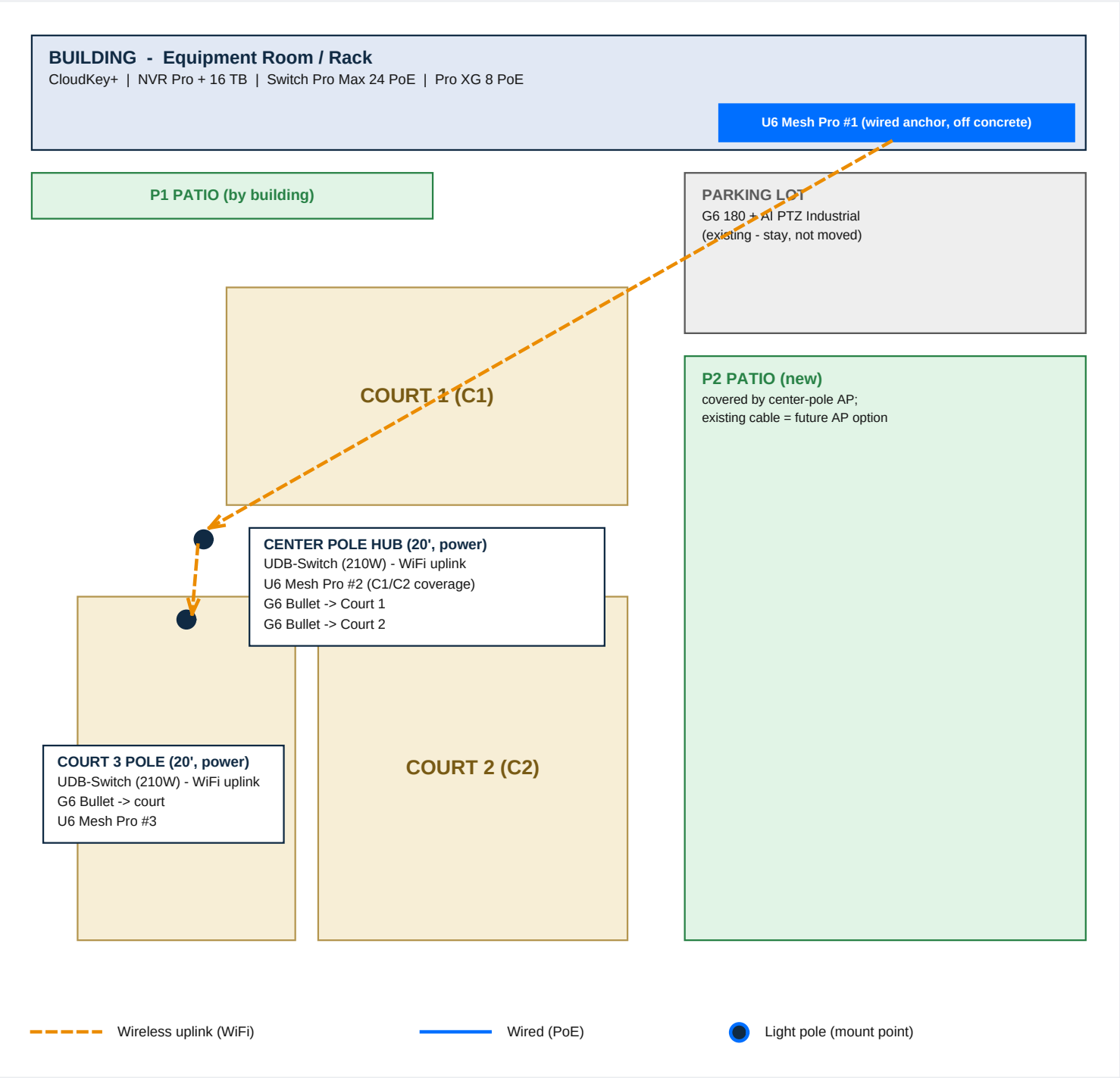
Item	Qty	Role in this project
Network Video Recorder Pro + 16 TB HDD	1	Records all court cameras
CloudKey+ (SSD)	1	UniFi Network/Protect controller
Switch Pro Max 24 PoE / Pro XG 8 PoE	2 / 1	Rack; powers wired building AP
Access Point U6 Mesh Pro	3*	*Relocate 2 off wall + add 1 new = 3 on poles
AI PTZ Industrial (IP66, -40C)	1	STAYS on parking lot (not moved)
Camera G6 180	1	STAYS on parking lot (do NOT use for courts)
Switch Flex 2.5G PoE + 210W adapter	1	Not needed at poles (UDB-Switch replaces it)

Confirmed site facts

- Center pole: new 20 ft pole (2 lights) between Court 1 and Court 2 - POWER PRESENT.
- Court 3 pole: 20 ft light pole - POWER PRESENT.
- P2 patio post: existing Cat6 pulled to it but NO power (kept as a future AP option only).
- Rack is inside the building, ~130 ft from the center-pole location.
- Courts each ~29'6" x 59'1" (approx 30 x 60 ft); perimeter light poles ~20 ft.
- Backhaul is wireless: a WiFi-7 UDB-Switch uplinking to a WiFi-6 U6 Mesh Pro links at WiFi-6 rates (hundreds of Mbps)
- far above the ~50-80 Mbps the cameras need.

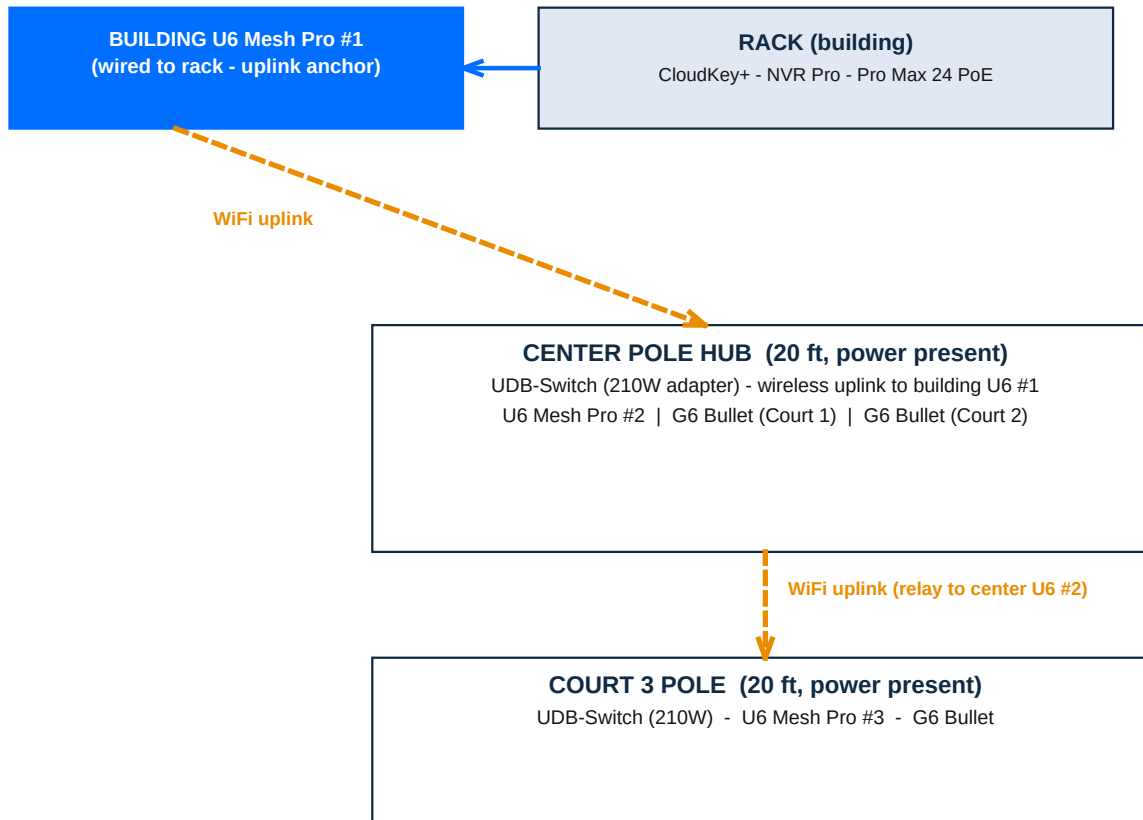
2. Site Layout & Device Placement

Schematic - relative positions per site photos/CAD; not to exact scale. Solid = wired (PoE). Dashed = wireless uplink (WiFi).



Orientation per owner: building at top with rack; P1 patio adjacent to building; Court 1 nearer building, Court 2 further; Court 3 to the left behind Court 2; P2 (new patio) along the right.

3. Network Topology & Wireless Uplinks



How the wireless backhaul works

- The UDB-Switch is a WiFi-7 switch with a built-in wireless uplink: it joins the UniFi WiFi over the air and provides PoE locally - so no cable runs out to the poles.
- Center pole UDB-Switch uplinks to the building U6 Mesh Pro #1 (~130 ft, line-of-sight).
- Court 3 UDB-Switch uplinks (relay) to the center-pole U6 Mesh Pro #2 - measured 110 ft, well within range (Court 3 sits behind Court 2, so it chains through the elevated center pole).
- WiFi-7 uplink to a WiFi-6 AP runs at WiFi-6 rates - hundreds of Mbps, far above the ~50-80 Mbps needed for 3x 4K cameras + AP clients.
- Each UDB-Switch is powered by a 210W adapter (up to 185W PoE) - ample for cameras + AP.
- Position the building U6 with clear line-of-sight to the center pole (mast/standoff off the concrete) so the uplink is strong.

4. Device-by-Device Placement & Mounting

Location	Device	Placement / Aim	Height	Power & Data
Building	U6 Mesh Pro #1 (relocated)	Mast/standoff OFF concrete; LoS to center pole	Roofline	Wired PoE from rack switch
Center pole	UDB-Switch (in NEMA enclosure)	Wireless uplink to building U6 #1	~6 ft (reach)	AC from pole light circuit + 210W adapter
Center pole	G6 Bullet	Aim down Court 1	~18-20 ft	PoE from UDB-Switch
Center pole	G6 Bullet	Aim down Court 2	~18-20 ft	PoE from UDB-Switch
Center pole	U6 Mesh Pro #2 (relocated)	Radiate over C1/C2; reaches P1/P2	~15-18 ft	PoE from UDB-Switch
Court 3 pole	UDB-Switch (in NEMA enclosure)	Wireless uplink to center-pole U6 #2	~6 ft (reach)	AC from pole light circuit + 210W adapter
Court 3 pole	G6 Bullet	Aim down length of Court 3	~18-20 ft	PoE from UDB-Switch
Court 3 pole	U6 Mesh Pro #3 (new)	Radiate over Court 3	~15-18 ft	PoE from UDB-Switch
Patio (pole)	G6 PTZ (new)	Pole-mount; sight line to 2nd patio; zoom/auto-track	on pole	Wired PoE+ (reuse 180 drop)
Main patio	G6 Bullet (new)	Back wall, LEFT - covers main patio	~10-12 ft	Wired PoE from building
Main patio	G6 Bullet (new)	Back wall, RIGHT - covers main patio	~10-12 ft	Wired PoE from building
Parking lot	G6 180 + AI PTZ Industrial	No change - stay on lot	existing	existing

Camera aiming: mount each bullet at one END of its court and aim down the 60 ft length for full coverage and best pixel density. Do not use the G6 180 for court detail. (A G6 Pro Bullet could be added later on a court for varifocal zoom / special-games detail.)

5. Bill of Materials

New hardware to purchase

Item (UniFi SKU)	Qty	Unit	Line
Device Bridge Switch (UDB-Switch-US)	2	\$299	\$598
AC Adapter 210W (UACC-Adapter-AC-210W)	2	\$79	\$158
Access Point U6 Mesh Pro (new; + relocate 2 owned)	1	\$179	\$179
Camera G6 Bullet (UVC-G6-Bullet-B) - one per court	3	\$199	\$597
Outdoor NEMA enclosure sized for UDB-Switch	2	~\$90	~\$180
Ethernet Surge Protection Outdoor (UACC-ETH-SP-Pro)	2-4	\$19	\$38-76
Pole mounts / building mast + short patch cables	1 lot	~\$120	~\$120

Estimated net new spend: approx. \$1,850 - \$1,950 (+ bar-TV viewer, below)

Cameras: 3x G6 Bullet selected (one per court) for full coverage. A G6 Pro Bullet (varifocal) can be added later on a court for zoom detail / special-games footage. Tariff and memory surcharges from the UniFi store are extra.

Patio security cameras (added)

Item (UniFi SKU)	Qty	Unit	Line
Camera G6 PTZ (UVC-G6-PTZ) - pole; 2nd patio + zoom	1	\$399	\$399
Camera G6 Bullet (UVC-G6-Bullet) - 2 on main-patio wall	2	\$199	\$398

Patio cameras subtotal: approx. \$797 (saved ~\$560 vs G6 Pro Bullets)

All three patio cameras wire directly to rack PoE+ (the 2 bullets from the building back wall; the PTZ reuses the current 180 drop) - NOT through the court UDB-Switch backhaul. Relocate or retire the patio-facing G6 180 (no zoom). The 2 G6 Bullets cover the MAIN patio (mounted either side of the back wall); the G6 PTZ provides zoom/auto-track and the sight line to the 2nd patio. If you later want continuous fixed coverage of the 2nd patio, add one more G6 Bullet there.

Customer viewing - bar TV (UniFi Protect on a screen)

- Existing Protect Viewport is in use in the office (all-camera wall) - not available for the bar.
- In stock now: Apple TV 4K (~\$129) + the official UniFi Protect tvOS app - shows a live court view on the bar TV. Use the 3rd-party Streamie app if you want a curated courts-only layout.
- Or buy a 2nd Protect Viewport when back in stock (native, full custom Live View support).
- In Protect, build a Live View with the 3 court bullets (audio muted), set as default, leave on.

Removed from earlier cart (not needed)

- U7 Outdoor APs x2 - not needed; U6 Mesh Pro is already omnidirectional and outdoor-rated.
- Switch Flex Utility x1 - that is an IP65 enclosure for a Flex SWITCH, not the larger UDB-Switch; use UDB-sized NEMA enclosures instead.
- Extra 2 cameras (cart had 5; design needs 3).

Reused (already owned - \$0)

- 3x U6 Mesh Pro APs (relocate 2 off building wall, add the 1 new = 3 on poles)
- Network Video Recorder Pro + 16 TB HDD - recording
- CloudKey+ controller; rack PoE switches power the wired building AP
- AI PTZ Industrial + G6 180 - stay on parking lot

6. Installation & Configuration

Physical install sequence

1. Re-mount a U6 Mesh Pro on the building OFF the concrete (short mast/standoff at the roofline) with clear line-of-sight to the center pole; wire it to a rack PoE port. This is the uplink anchor.
2. Mount a NEMA enclosure on the center pole; install the UDB-Switch + 210W adapter; have an electrician land pole-light-circuit power to it (no trenching - power is at the pole).
3. Mount center-pole devices: G6 Pro Bullet (main court), G6 Bullet (other court), U6 Mesh Pro #2. Patch all to the UDB-Switch PoE ports.
4. Repeat at Court 3: NEMA enclosure + UDB-Switch + 210W adapter (pole power), G6 Bullet, U6 Mesh Pro #3.
5. Install ETH-SP surge protectors on the wired building-AP run and any cable entering the building.
6. Aim cameras down each court; tilt APs slightly down toward the sand.

UniFi configuration

- Adopt the building U6 (wired) first; build one outdoor WLAN; enable band steering.
- Adopt each UDB-Switch and set its wireless uplink to the correct AP (center -> building U6; Court 3 -> center U6). Confirm uplink RSSI better than -60 dBm and a healthy link rate.
- Adopt the pole APs (U6 #2, #3). Set fixed, non-overlapping 5 GHz channels across the 3 APs; reduce TX power from High to Medium once on poles; verify with a follow-up heat map.
- Adopt cameras in UniFi Protect; confirm recording to NVR Pro; set AI/motion zones to the courts; set the G6 Pro Bullet zoom to fill the main court.
- Confirm each UDB-Switch PoE load is within budget (185W with the 210W adapter).

PoE budget per pole (UDB-Switch, 185W available with 210W adapter)

Pole load	Approx draw
Center pole: 2x G6 Bullet + U6 Mesh Pro	~27 W
Court 3: G6 Bullet + U6 Mesh Pro	~20 W
Headroom on 185W budget	very large

7. Distances, Cabling & Power Schedule

Court/area dimensions are from the site CAD. Link distances marked EST. are approximate - measure pole-to-pole on site (laser or wheel) and record in the right column before ordering cable. Confirm clear line-of-sight at mount height for each wireless uplink.

Court & area dimensions (from CAD)

Feature	Dimension
Court 1 / 2 / 3 playing surface (each)	29'6" x 59'1" (~30 x 60 ft)
Courts 1 & 2 fenced area	100'0" x 90'0"
Court 3 fenced area	61'0" x 89'6"
Perimeter light poles	~20 ft
Center pole (new, 2 lights)	20 ft - power present

Link & cable-run schedule

Segment	Type	Est. distance	Measured
Rack -> building U6 #1	Wired Cat6	In building (short)	_____
Building U6 #1 -> center pole	WIRELESS uplink	~130 ft (owner)	_____
Center pole -> Court 3 pole	WIRELESS relay	110 ft (measured)	DONE
Center pole UDB -> each camera	Wired patch	~15-30 ft	_____
Center pole UDB -> U6 #2	Wired patch	~10-20 ft	_____
Court 3 UDB -> camera	Wired patch	~15-30 ft	_____
Court 3 UDB -> U6 #3	Wired patch	~10-20 ft	_____

Power schedule (electrician)

- Center pole: tap existing pole light circuit (120V) -> 210W adapter (in enclosure) -> UDB-Switch.
- Court 3 pole: tap existing pole light circuit (120V) -> 210W adapter (in enclosure) -> UDB-Switch.
- Building U6 #1: no separate power - PoE from a rack switch port.
- Cold note: the 210W adapter is rated 0-40 C (32-104 F). Courts are seasonal, but if the gear stays powered through deep winter use a sealed enclosure (gear self-heat helps) and monitor.

Cable & accessory takeoff

- Outdoor/UV or direct-burial shielded Cat6 for the pole patch runs + the building-AP run.
- ETH-SP-Pro surge protector on the building-AP copper run (and any copper entering a structure).
- Stainless band-clamp pole mounts (3 cameras + 2 pole APs); building mast/standoff for U6 #1.
- Weatherproof glands, drip loops, UV zip ties, dielectric grease, outdoor RJ45 ends + tester.

8. Commissioning & Acceptance Checklist

Wireless backhaul (UDB-Switch uplinks)

- ☐ Center-pole UDB-Switch uplink online; RSSI better than -60 dBm; link rate healthy.
- ☐ Court 3 UDB-Switch uplink (relay) online; all Court 3 devices reachable.
- ☐ Pull a phone/laptop speed test through each pole - well above camera needs.

WiFi coverage (re-survey)

- ☐ Re-run the WiFiman heat map across all 3 courts + both patios.
- ☐ Court surface now reads -55 dBm or better (was -70 to -80).
- ☐ No dead pocket on P1/P2 (existing P2 cable can feed an added AP if needed).
- ☐ Roaming: a phone walks court-to-court without dropping.

Cameras (UniFi Protect)

- ☐ All 3 court bullets online and recording to NVR Pro.
- ☐ Each bullet frames its full court end-to-end.
- ☐ Bar TV: Apple TV (or Viewport) shows the courts Live View; audio muted; always-on.
- ☐ Night check: image usable under the Musco court lights.
- ☐ Retention check: 16 TB HDD holds target days with the added cameras (add HDD if short).

Power & weatherproofing

- ☐ UDB-Switch + 210W adapter sealed inside NEMA enclosures; cable glands tight; drip loops.
- ☐ ETH-SP surge protectors grounded on the wired building run.
- ☐ No PoE over-budget warnings on either UDB-Switch.

Sign-off

Installer: _____ Date: _____

Notes: _____

9. Device Specifications (Appendix)

Key manufacturer specs for each device. Verify current values on ui.com before purchase.

Device Bridge Switch (UDB-Switch-US) - NEW x2

Ports	1x 10 GbE + 7x 2.5 GbE; PoE+ output
PoE budget	35W (incl. 60W adapter) / up to 185W (with 210W adapter)
Wireless uplink	WiFi 7, 4 streams; up to 5.8 Gbps (6 GHz) / 4.3 Gbps (5 GHz)
Power	54V DC; 60W adapter included, 210W optional
Size / class	212.9 x 113 x 32.5 mm; desktop/wall (indoor-class - enclose outdoors)

Access Point U6 Mesh Pro (U6-MESH-PRO) - REUSE x2 + NEW x1

WiFi	WiFi 6; 5 GHz 2x2 to 2.4 Gbps, 2.4 GHz 2x2 to 573.5 Mbps; 8 dBi
Ports / power	2x GbE RJ45; PoE (48V adapter incl. or PoE switch); 9W max
Environment	IPX6 weatherproof; -30 to 60 C; 250+ clients
Size	343.2 x 181.2 x 60.2 mm

Camera G6 Bullet (UVC-G6-Bullet-B) - NEW x5 (3 courts + 2 patio)

Imaging	8MP 4K (3840 x 2160), 1/1.8" sensor, 30 FPS, AI detection
Field of view	109.9 H / 56.7 V / 134.1 D (degrees)
Night vision	IR up to 30 m (98 ft); IR cut filter
Power	PoE 37-57V DC; 9.9W max
Environment	IP66; -20 to 50 C

Camera G6 PTZ (UVC-G6-PTZ) - NEW x1 (patio)

Imaging	4K, 1/1.8" 8MP sensor, 30 FPS; dual-lens (wide + telephoto)
Zoom / motion	10x hybrid (4x optical + 2.5x digital); 350 pan / 100 tilt
AI	Auto-tracking of people/vehicles; full smart detections
Power / env	PoE+; outdoor-rated IP dome; approx. \$399

AC Adapter 210W (UACC-Adapter-AC-210W) - NEW x2

Input / output	100-240V AC in; 54V DC out, 3.90A, 210W
Operating temp	0 to 40 C (32-104 F) - cold-weather caveat
Size	190 x 86 x 26 mm; C14 inlet, 1.5 m DC cable

Ethernet Surge Protection Outdoor (UACC-ETH-SP-Pro) - NEW x2-4

Protection	Passive, 20kA discharge; data to 10 Gbps; 60W PoE++ passthrough
Environment	IPX5; -30 to 65 C
Size	91 x 61 x 32.5 mm

Network Video Recorder Pro (UNVR-Pro) - EXISTING

Storage	7x 2.5/3.5" bays; SATA to 24TB each; up to 168TB raw; RAID
Capacity	Up to 24x 4K cameras (~60 days) or 70 HD
Ports / form	1x 10G SFP+, 1x GbE RJ45; 2U rackmount

Apple TV 4K (bar viewer) - NEW x1 (alternative to Viewport)

Use	Runs official UniFi Protect tvOS app; HDMI to bar TV
Network	Ethernet (higher model) or WiFi; approx. \$129